Abdullah Al Miraj

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LinkedIn | GitHub | Website

Research Interests

Flexible and Printed Electronics; Biomedical Devices and Sensors; Semiconductor Device Fabrication

Education

Ohio University *Fall 2025–Present*
Master of Science in Electrical Engineering Athens, Ohio, USA

- CGPA: 3.90/4.00 after the first three semesters.

Chittagong University of Engineering & Technology *June 2022*
Bachelor of Science in Electrical and Electronic Engineering Chittagong, Bangladesh

- CGPA: 3.18/4.00.

Published & Ongoing Research

1. **Abdullah Al Miraj**, Grace Sunday, Sumaiya Abul Kalam, and Savas Kaya, "Ultra-High-Resolution Copper-Coated LIG Sensors via Electroless Deposition on UV-Scribed Kapton," IEEE SENSORS 2026, Netherlands.
2. Sumaiya Abul Kalam, **Abdullah Al Miraj**, Grace Sunday, Weiguang Yuan, and Savas Kaya, "High-Resolution Laser-Induced Graphitization on Kapton: Case for Thermal Control and PMMA Cap," IEEE FLEPS 2026, USA. doi:10.1109/FLEPS69128.2026.11645038
3. Sumaiya Abul Kalam, **Abdullah Al Miraj**, Weiqiang Yuan, and Savas Kaya, "Toward High Resolution Laser Induced Graphitic Structures: Wavelength, PMMA Capping, and Polymer Dependent Processing," IEEE Journal on Flexible Electronics, **Submitted Journal Manuscript**.
4. **Abdullah Al Miraj**, Asif Mahbub Uddin, Md. Manjurul Gani, Tasnia Sultana, and Sharmin Shultana, "Performance Analysis of Different Piezoelectric and Shim Materials on Bimorph Piezoelectric Energy Harvester," 2024. doi:10.1109/I2CT61223.2024.10543375
5. Asif Mahbub Uddin, **Abdullah Al Miraj**, Moumita Sen Sarma, Avishek Das, and Md. Manjurul Gani, "Traditional Bengali Food Classification Using Convolutional Neural Network," 2021.doi:10.1109/TENSYMP52854.2021.9550874 .
6. Tasnia Sultana, Manjurul Gani, Sharmin Shultana, **Abdullah Al Miraj**, Asif Mahbub Uddin, and Joyprokash Chakrabartty, "Effect of Scandium Concentration on the Performances of Cantilever-Based AlN Unimorph Piezoelectric Energy Harvester with Silicon Nitride Substrate," 2024. doi:10.1007/s40243-024-00272-9 .
7. Reshad Ibn Mumin, Raxit Ridon, Umme Salma Muna, and **Abdullah Al Miraj**, "Design and Fabrication of Bengali Braille Embosser Using Solenoid," 2024. doi:10.1109/I2CT61223.2024.10543990 .

Research and Technical Experience

Micro/Nanofabrication, Characterization, and Imaging

- **Laser-Induced Graphene:** Fabrication and patterning using UV, IR, and blue laser systems on flexible polymer substrates including PI and PET.
- **Sensor Fabrication and Characterization:** Fabrication of LIG-based capacitive sensors on different polymer substrates; capacitance and dynamic-response characterization using the Texas Instruments Capacitive Reader SV601093, Analog Discovery 2, LCR meters, and related measurement tools.
- **Electrical Characterization:** I–V, impedance, resistance, capacitance, and inductance measurements using SMUs, Agilent E4980A Precision LCR Meter, and two-/four-probe methods.
- **Micro/Nanofabrication:** Spin coating, thin-film preparation, substrate processing, and basic lithography process development.
- **Lithography:** Mask-based photolithography using the Karl Suss MJB3 Aligner and maskless patterning using the POLOS Beam Optical Writer, including process troubleshooting.
- **Microscopy:** SEM imaging using JEOL JSM-6490 and Phenom G2 Pro, optical microscopy, and AFM characterization using Nanosurf NaioAFM.

Teaching Experience

Ohio University, EECS *Fall 2026–Present*
Graduate Teaching Assistant, ET 2100 – Engineering Programming Athens, Ohio, USA

Assist undergraduate students with Python programming, debugging, engineering programming assignments, and development environment setup using VS Code.

Professional Experience

Renata PLC, Bangladesh *July 2022–July 2025*
Assistant Manager, Engineering Department Mirpur, Bangladesh

Led a nine-member engineering team, supervised preventive and breakdown maintenance, troubleshot industrial automation systems, coordinated electrical layouts and switchgear installation, supported diesel-generator and substation operations, monitored HVAC systems for ISO Class 8 and critical production areas, and maintained the purified water system.

Selected Academic and Engineering Projects

Real-Time Sound Classification and Notification System using Arduino Nano 33 BLE Sense Rev2

- Built an end-to-end embedded system that recognizes everyday sounds in real time and transmits the detected class to a smartphone through Bluetooth Low Energy (BLE), integrating embedded sensing, on-device machine learning, and wireless communication.

Light-Intensity-Based Motor Direction Control

- Developed an H-bridge-based motor control system responsive to light intensity, with the objective of shifting selected agricultural loads away from daytime peak-demand periods.

Software and Programming Skills

Programming	Python, C
Scientific and Engineering Tools	MATLAB, Cadence, Jupyter Notebook
Documentation	LaTeX, Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Sublime Text

Training and Certifications

Huawei Seeds for the Future Programme *September 7–11, 2020*
Trainee

- Completed training in cloud computing, 5G, artificial intelligence, and Huawei Mobile Services.

Training Institute for Chemical Industries *January 11–31, 2020*
Industrial Technology in Electrical Engineering and Instrumentation Palash, Narsingdi, Bangladesh

English Proficiency

Duolingo English Test(Taken on July 2026): Overall 140 (Speaking 135, Writing 150, Reading 140, Listening 135)

Honors and Awards

- First Runner-Up, Review Paper Segment, COVID-19 Combatants Unification Competition, IEEE BRAC Student Branch, 2020.
- First Runner-Up, Project Show, TechnoCraze 2019, Andromeda Space Research & Robotics Organization.
- CLAP Challenge Champion, IEEE Region 10 Young Professionals.

Leadership and Service

IEEE Student Branch, CUET *November 2020–May 2022*
General Secretary

Andromeda Space & Robotics Research Organization, CUET *December 2020–September 2022*
Vice President